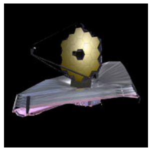

12340 - Detecting the Earliest Galaxy Overdensity within the First 500 Myr

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	GS-z11-overdensity	NIRSpec MultiObject Spectroscopy	(1) overdensity_members

ABSTRACT

Understanding the emergence of the first large-scale structures is crucial for early galaxy formation theory and cosmological models. Despite the remarkable number of galaxies discovered at z > 10, none have yet been observed within a significant overdensity at such early epochs. This proposal will confirm the first candidate z = 11 galaxy overdensity in the GOODS-S field. Using JWST/NIRSpec PRISM and medium-resolution gratings, we will obtain spectroscopic redshifts for member galaxies, search for Lya emission, and measure rest-UV emission lines to constrain metallicity, ionization conditions, and potential black hole activity in an overdensity environment. These data will directly probe how dense environments influence early galaxy evolution, star-formation efficiency, and the production of ionizing photons that drive cosmic reionization.

We will use NIRSpec PRISM observations in the MSA mode to spectroscopically confirm member galaxies and NIRSpec medium gratings to measure narrow emission lines by stacking the spectra. From these data, we will derive metallicities and Lya line strengths to study the ionization state of the intergalactic medium and the progression of reionization. We expect to confirm 11 members within the overdensity, providing the first robustly verified z > 10 proto-cluster and delivering critical constraints on early structure assembly and galaxy formation in overdensity environments.

OBSERVING DESCRIPTION

We are conducting MSA spectroscopy of an overdensity of z=11 candidates, seeking to confirm the Lya dropout and searching for Lya emission.

These objects are faint, as low as 8 nJy in the continuum, and we are searching for lines to 20 A rest-EW, which is 2.7e-19 cgs line flux. This calls for 50 ksec with the PRISM to reach SNR = 7 per 9 pixels (0.1 micron) in the continuum, and 50 ksec with G140M/F100LP to reach SNR = 5 on the line detection. It is important to use the grating to isolate the line; otherwise, the line is not detected and the possibility of its presence causes degeneracies in the fitting of neutral hydrogen damping wings.

We build using integrations of 19 groups with NRSIRS2. We use 3-shutter nods, 3 integrations per location, and then use 4 nods in each disperser to reach 50 ksec.

We are then conducting coordinated parallels with NIRCam to create a 13-filter deep field just outside of the current GOODS-S footprint, designed to extend the search for z=11 galaxies. This involves 3 integrations of 7 groups of DEEP8 in each location. Each filter will only have 3 dither positions, but each position has at least 3 integrations with which to reject detector outliers and cosmic rays. F115W is observed to triple-depth, as it is the z=11 dropout filter. F480M is observed to double depth as it is noisier and yet important as to the one filter redward of the z=11 Balmer break.

Proposal 12340 - Targets - Detecting the Earliest Galaxy Overdensity within the First 500 Myr

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	overdensity_members	RA: 03 31 52.8012 (52.9700050d) Dec: -27 48 28.14 (-27.80782d) Equinox: J2000		

Comments:
Description=[]

Observation

Proposal 12340, Observation 1: GS-z11-overdensity

Fri Mar 13 23:03:18 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Coordinated Parallel Template(s): NIRCam Imaging

Diagnostics

(GS-z11-overdensity (Obs 1)) Warning (Form): Line 3 (G140M/F070LP) is currently unused in any Configuration/Pointing.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	overdensity_members	RA: 03 31 52.8012 (52.9700050d) Dec: -27 48 28.14 (-27.80782d) Equinox: J2000		

Comments:
Description=[]

Acquisition

NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

NIRSpec MultiObject Spectroscopy

TA Method: MSATA

HFF Readout Mode: false

Obtain Confirmation Images: No

Science Aperture: MSA Center

Primary Candidate List: overdensity_members (19 sources)

Filler Candidate List: null

Spectral Overlap Map: jwst-nirspec-hr

Spectral Overlap Threshold: 1.5

NIRCam Imaging

Module: ALL

Subarray: FULL

Reference Stars

Dithers

#	Dither Type
1	NONE

Proposal 12340 - Observation 1 - Detecting the Earliest Galaxy Overdensity within the First 500 Myr

Spectral Elements

NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	2 (G140M/F100LP)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
2	2 (G140M/F100LP)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
3	2 (G140M/F100LP)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
4	2 (G140M/F100LP)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
5	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
6	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
7	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801
8	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	52.98023 Degrees - 27.792172222222234 Degrees	99.99526711180874			3	9	12604.801

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F090W	F277W	DEEP8	7	3	9	3	12433.18	
2	F115W	F300M	DEEP8	7	3	9	3	12433.18	
3	F115W	F356W	DEEP8	7	3	9	3	12433.18	
4	F115W	F335M	DEEP8	7	3	9	3	12433.18	
5	F150W	F444W	DEEP8	7	3	9	3	12433.18	
6	F140M	F410M	DEEP8	7	3	9	3	12433.18	
7	F182M	F480M	DEEP8	7	3	9	3	12433.18	
8	F200W	F480M	DEEP8	7	3	9	3	12433.18	

Special Requirements	
	Group Visits within 53.0 Days Aperture PA Range 99 to 101 Degrees (V3 320.4254303 to 322.4254303) Visits Same PA No Parallel Attachments MSA Planned Aperture PA 100.0000 to 100.0000 Degrees (V3 321.4254303 to 321.4254303)